



Chartered Financial Data Scientist (CFDS)

Study on the Tail Distribution of Stock Returns: From Empirical Analysis to Factor Explanations

CFDS-11 – Project Presentation – Pierre Nibéron

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- Motivation
- Research Plan
- Data
- Part 1 - Description:
 - Whole S&P500 description
 - Tails description
 - Time clustering
 - Sector clustering
- Part 2 - Modeling:
 - CAPM
 - Fama-French 3 factors
 - Carhart's
 - Sentiment
- Conclusion

- On Fat-Tails
 - Mention by Dr. Sikandar Siddiqui during CFDS lessons
 - Personal IT Background going into finance
 - Describing and finding explanatory models
 - Reading literature on the subject
 - Using the tools learned during CFDS course
 - Initiation to existing financial models (CAPM, Fama-French, ...)

"large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes." – Benoit B. Mandelbrot, 1963

- First Part : Description and statistical testing
 - Do we observe fat tails which would not exist under normal assumptions ?
Arithmetic vs. Log, Skewness, Kurtosis, JB and visualization
 - What do these tails look like? Are they symmetrical?
99.9th percentile, clustering, t-test, f-test and visualization
- Second Part : Model and explanatory power
 - Do factor models explain extreme returns as well as they explain the full distribution?
CAPM, Fama-French Three factors model, Carhart's Momentum, Sentiment

- S&P500 stock list from wikipedia
- Daily stock prices from 01.01.2010 to 01.05.2026
- yFinance api (data from yahoo finance website)
- Risk free rate, T-Bill 3 month from FRED's website
- Fama-French dataset from panda_datareader library
- News headlines from kaggle (see link in sentiment analysis part)

```
date,Ticker,Open,High,Low,Close,Adj Close,Volume,previous_adj_close,arithmetic return,arithmetic return %,log_return,index,Dividends,Stock Splits,z_score global,extreme
2020-03-09,APA,13.420000076293944,13.699999809265137,9.31999969482422,9.550000190734863,8.224353790283203,28073200.0,17.826614379882812,-0.5386474618778808,-53.86474618778808,-0.7735928034616633,,,,-39.92385295519495,True
2024-04-11,GL,98.4800033569336,98.58000183105467,38.95000076293945,49.16999816894531,48.29851150512695,36577500.0,103.07022857666016,-0.5314019171966409,-53.14019171966409,-0.7580098442455365,,,,-39.1201694230346,True
2019-01-14,PCG,9.210000038146973,9.729999542236328,7.78000020980835,8.380000114440918,8.26025390625,127198800.0,17.338647842407227,-0.5235929594205784,-52.35929594205784,-0.7414826628960451,,,,-38.26778809421307,True
2020-03-09,OXY,15.579999923706056,19.190000534057617,12.039999961853027,12.510000228881836,11.708246231079102,104930300.0,24.39919090270996,-0.5201379308943112,-52.01379308943112,-0.7342565724046701,35302.0,0.79,0.0,-37.89510599646308,True
2025-10-29,FISV,71.36000061035156,76.6500015258789,66.58000183105469,70.5999984741211,70.5999984741211,103454200.0,126.16999816894533,-0.44043750892676065,-44.043750892676066,-0.5806000665408693,,,,-29.97034654228724,True
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2018-10-04,SMCI,1.5160000324249268,1.5160000324249268,0.8500000238418579,1.2599999904632568,1.2599999904632568,125601000.0,2.140000104904175,-0.41121498659006966,-41.12149865900697,-0.5296941646598585,,,,-27.34489944561519,True
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```

Arithmetic>Returns stats table:

```
count      1710528.000000
mean        0.000691
std         0.019362
variance    0.000375
min         -0.538647
1%          -0.052116
5%          -0.027356
25%         -0.007924
median      0.000739
75%         0.009405
95%         0.028037
99%         0.053622
max         0.745932
skewness    0.180127
kurtosis    23.823084
JB statistic 40458666.050451
JB p-value  0.000000
```



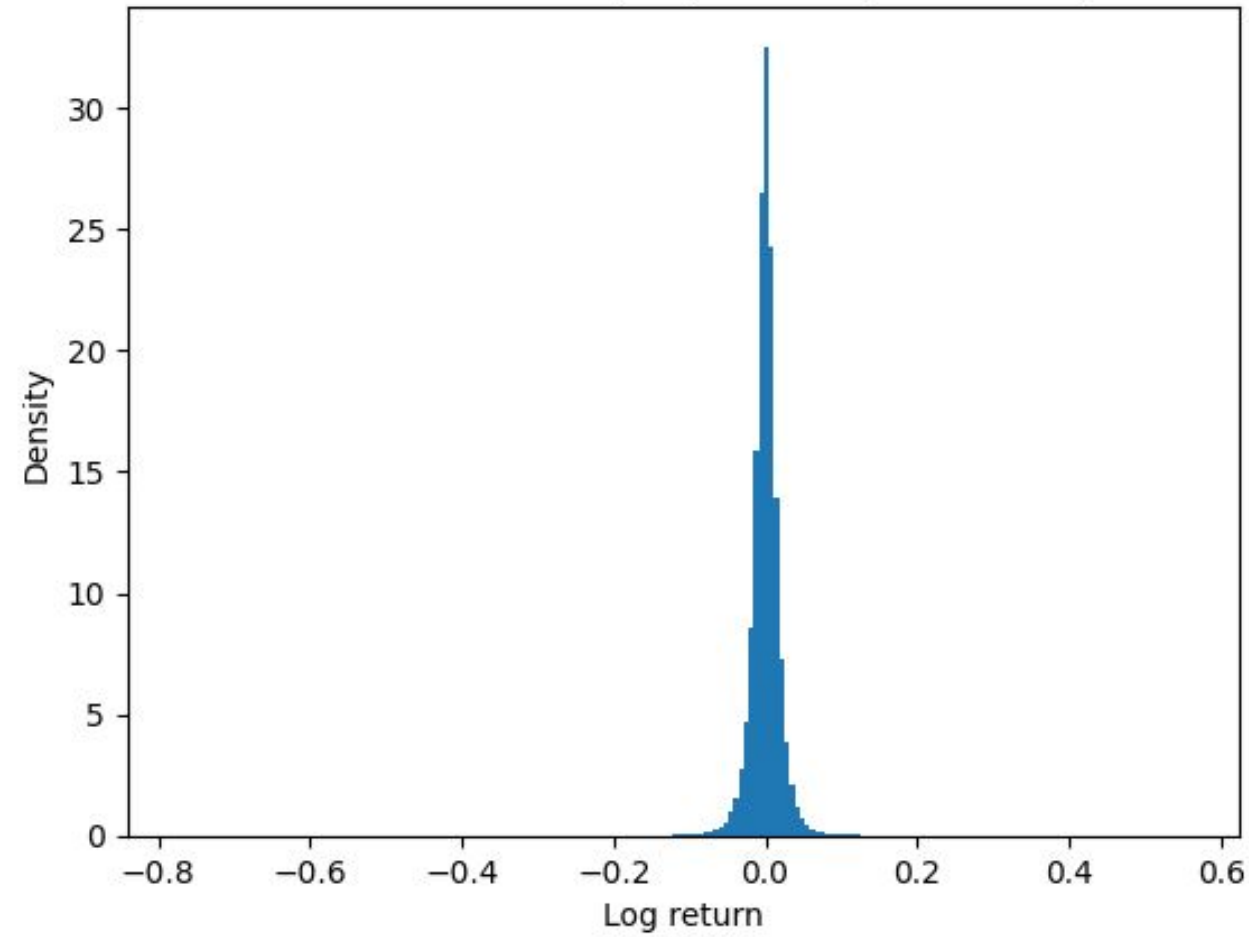
Log>Returns stats table:

```
count      1710528.000000
mean        0.000503
std         0.019403
variance    0.000376
min         -0.773593
1%          -0.053523
5%          -0.027737
25%         -0.007955
median      0.000738
75%         0.009361
95%         0.027652
99%         0.052233
max         0.557289
skewness    -0.580858
kurtosis    26.745270
JB statistic 51077412.921643
JB p-value  0.000000
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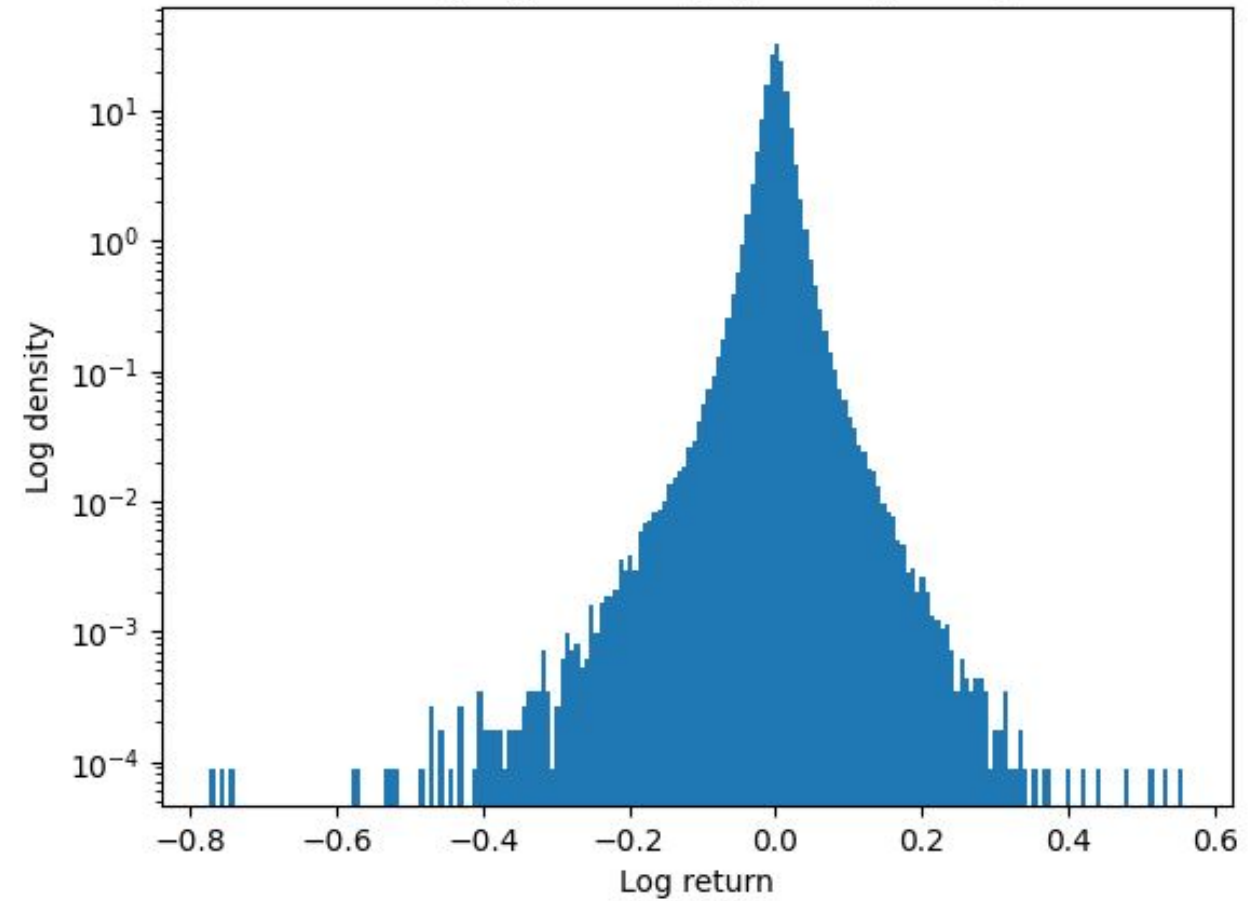


The log return definition, compared to the arithmetic one, flattens extreme positive returns and increase extreme negative returns

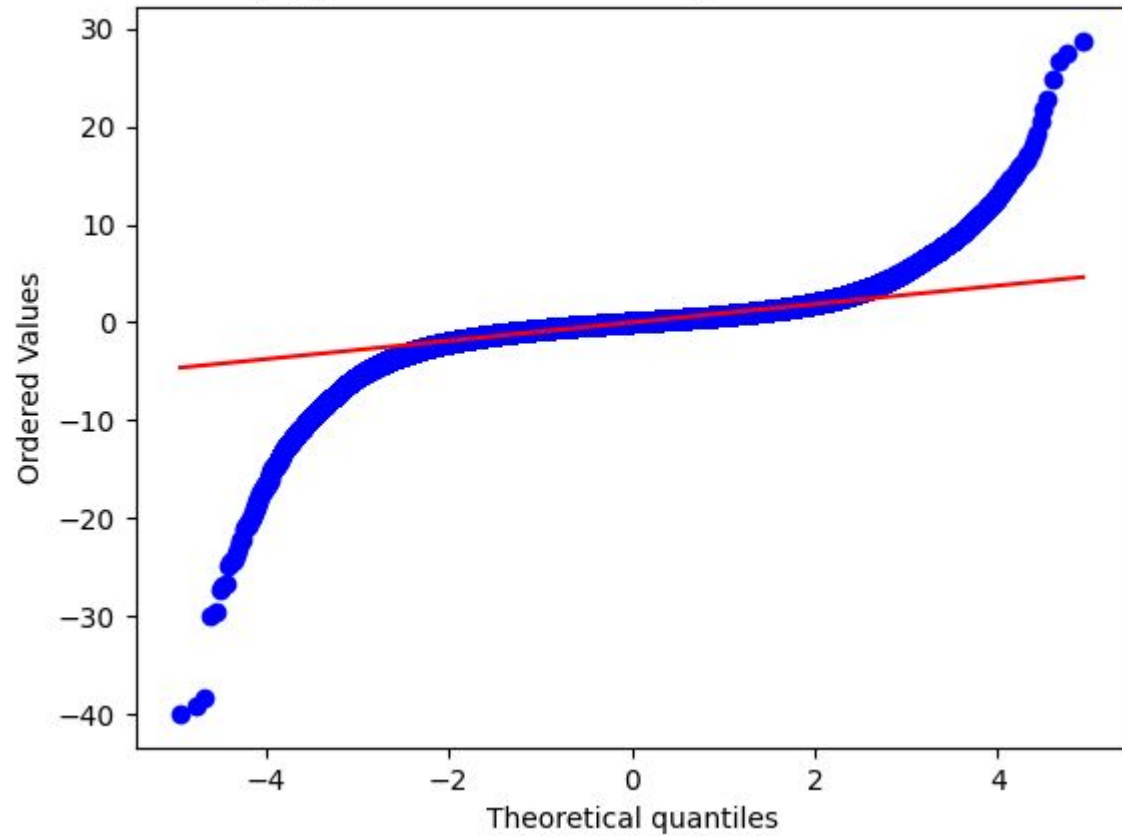
Distribution of daily log returns (linear scale)



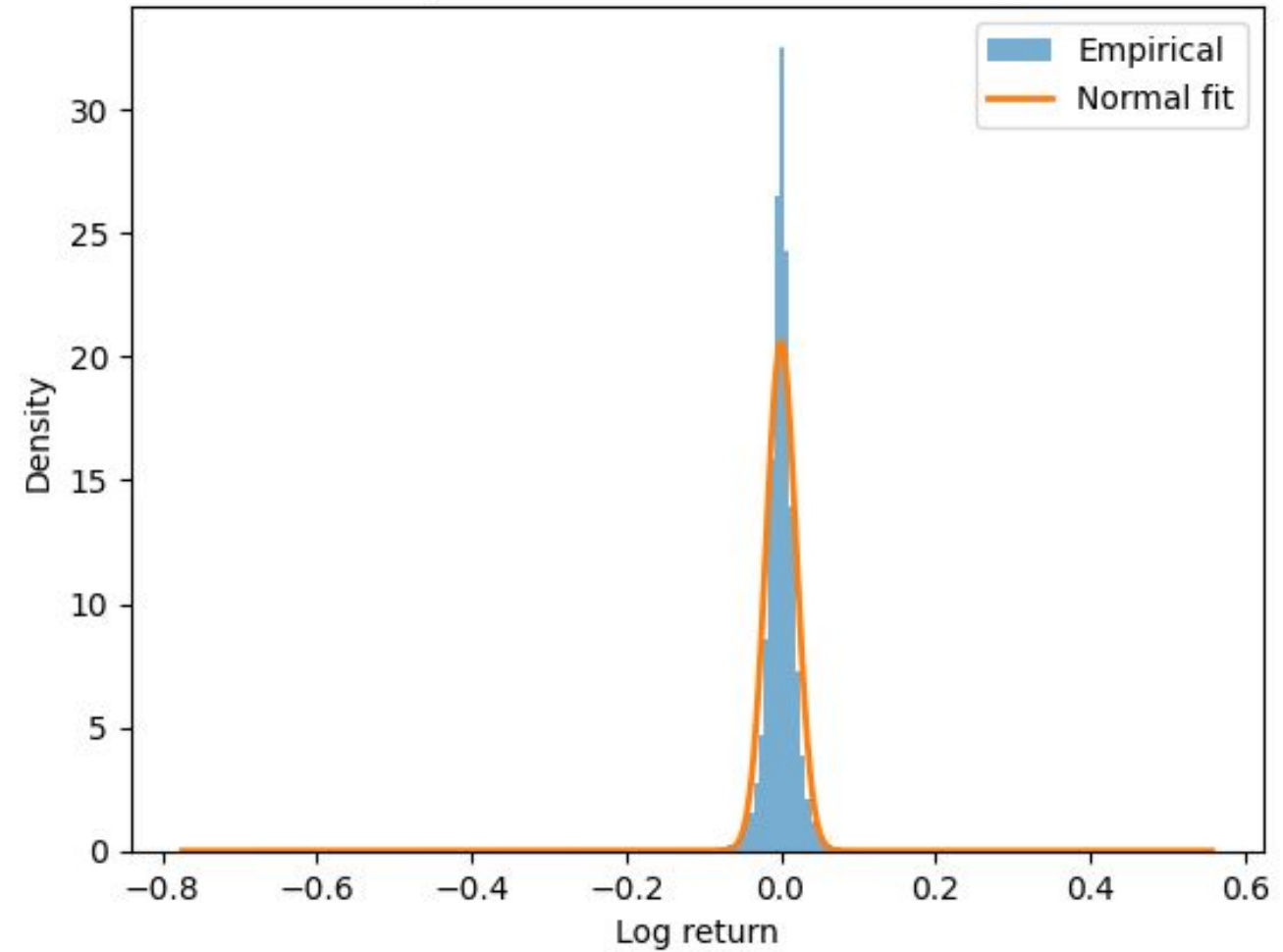
Daily log returns (log density scale)

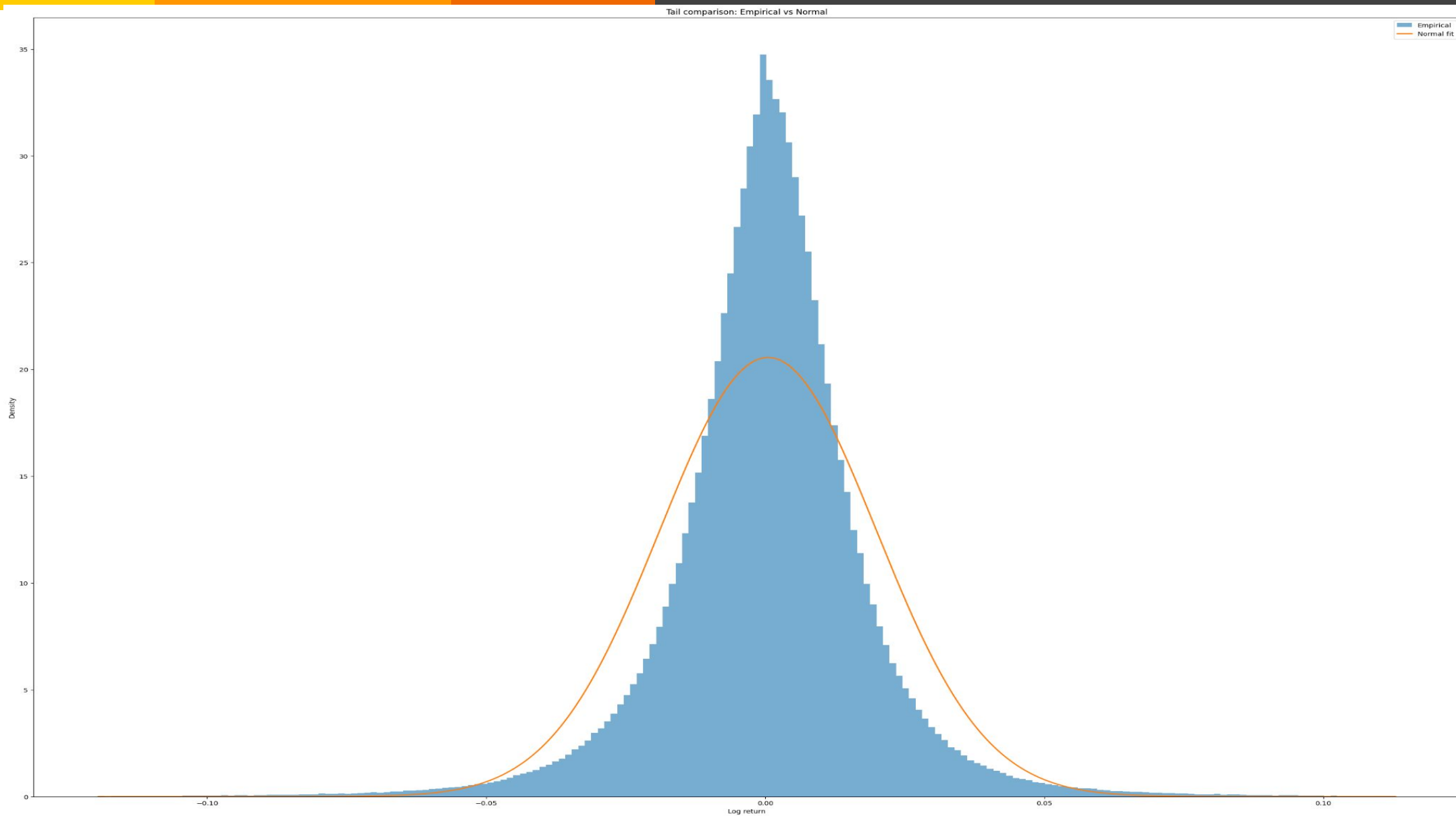


QQ-plot of standardized log returns vs Normal

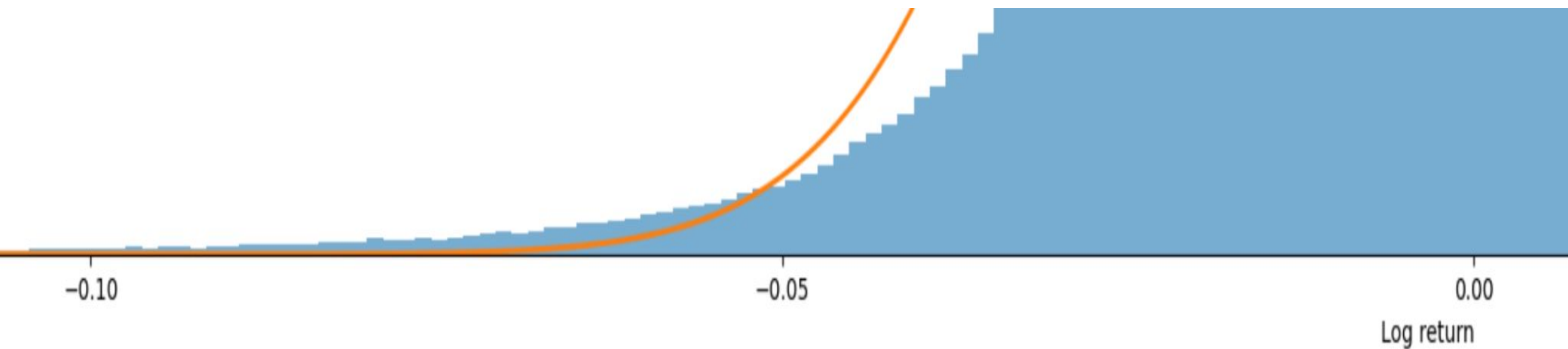


Log-returns vs Normal distribution

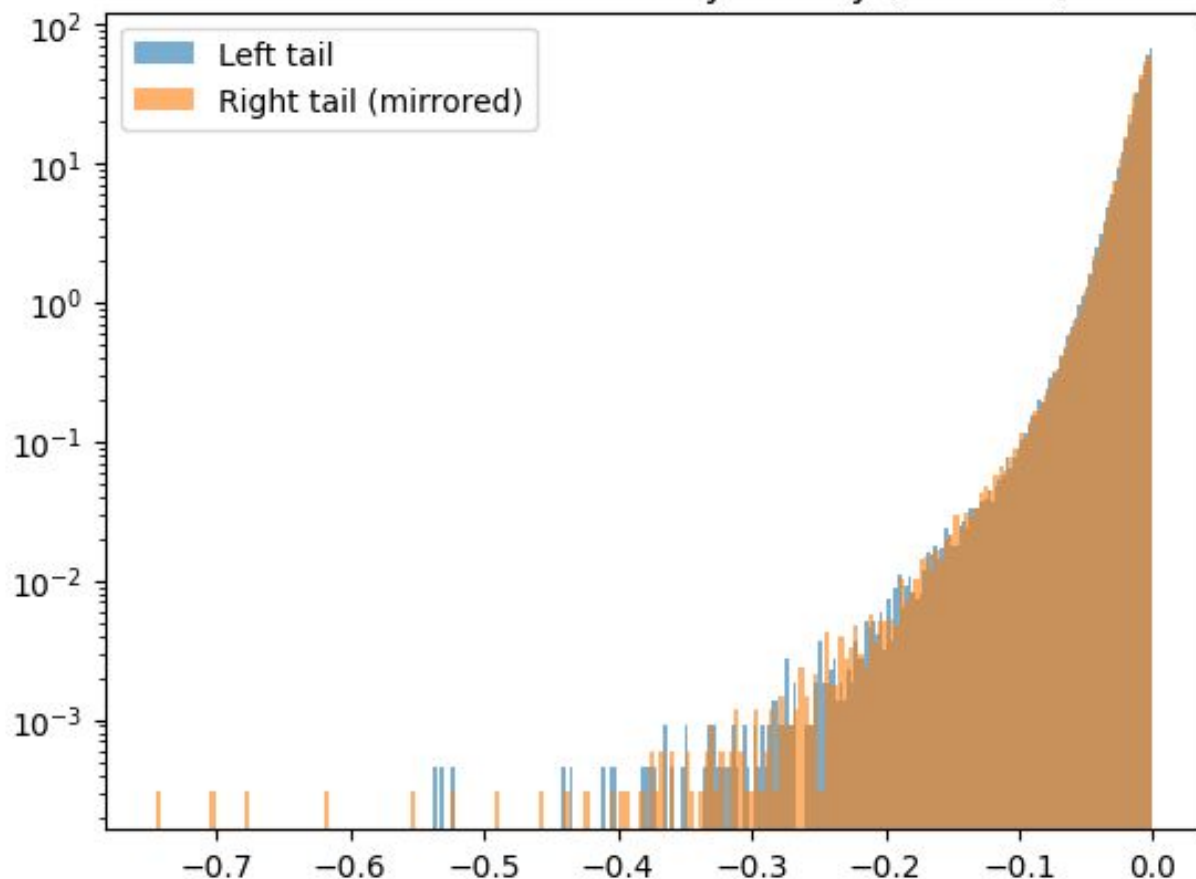




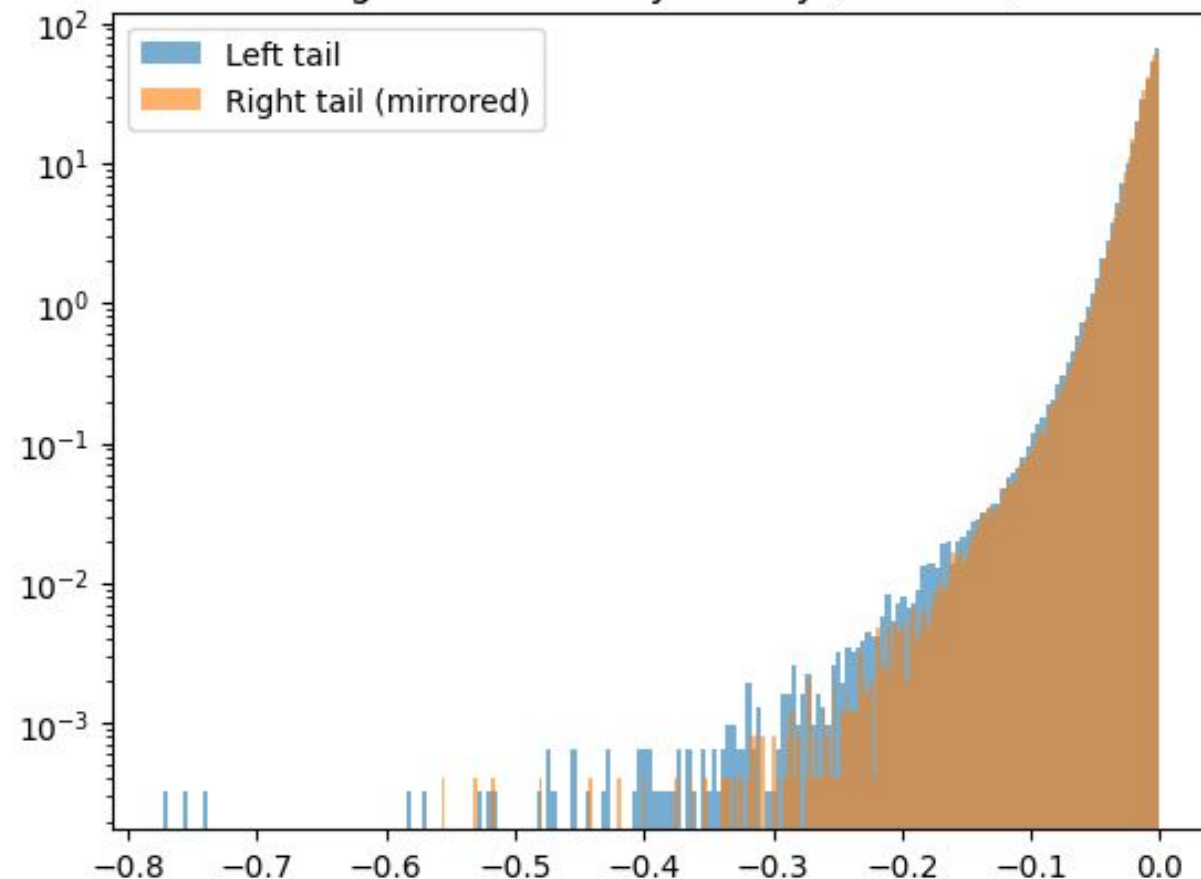
Whole set visualization



Arithmetic>Returns tail asymmetry (mirrored)



Log>Returns tail asymmetry (mirrored)



Again, we see the differences between return definitions in the extreme values.

- Do the two top 0.1% have the same statistical resemblance?
Check both return definition, arithmetic_return (left), log_return (right)

	positive_tail	negative_tail_abs
count	692.000000	1019.000000
mean	0.203309	0.177224
std	0.064695	0.050491
min	0.154012	0.133428
25%	0.166275	0.145062
50%	0.182088	0.161534
75%	0.218897	0.190182
max	0.745932	0.538647

	test	t_stat	p_value
0	Welch t-test	8.920870	1.628379e-18
1	Levene test	5.170664	2.309497e-02

	positive_tail	negative_tail_abs
count	692.000000	1019.000000
mean	0.183787	0.197229
std	0.049444	0.068200
min	0.143244	0.143210
25%	0.153815	0.156726
50%	0.167282	0.176181
75%	0.197947	0.210945
max	0.557289	0.773593

	test	t_stat	p_value
0	Welch t-test	-4.723884	0.000003
1	Levene test	13.545443	0.000240

More extreme negative events.

H0 rejected for equal means and variances between positive and negative extreme distributions

- Top 0.1% daily returns clustered by years

year	
2010	39
2011	73
2012	65
2013	51
2014	24
2015	53
2016	56
2017	37
2018	54
2019	61
2020	763
2021	20
2022	69
2023	51
2024	97
2025	176
2026	44

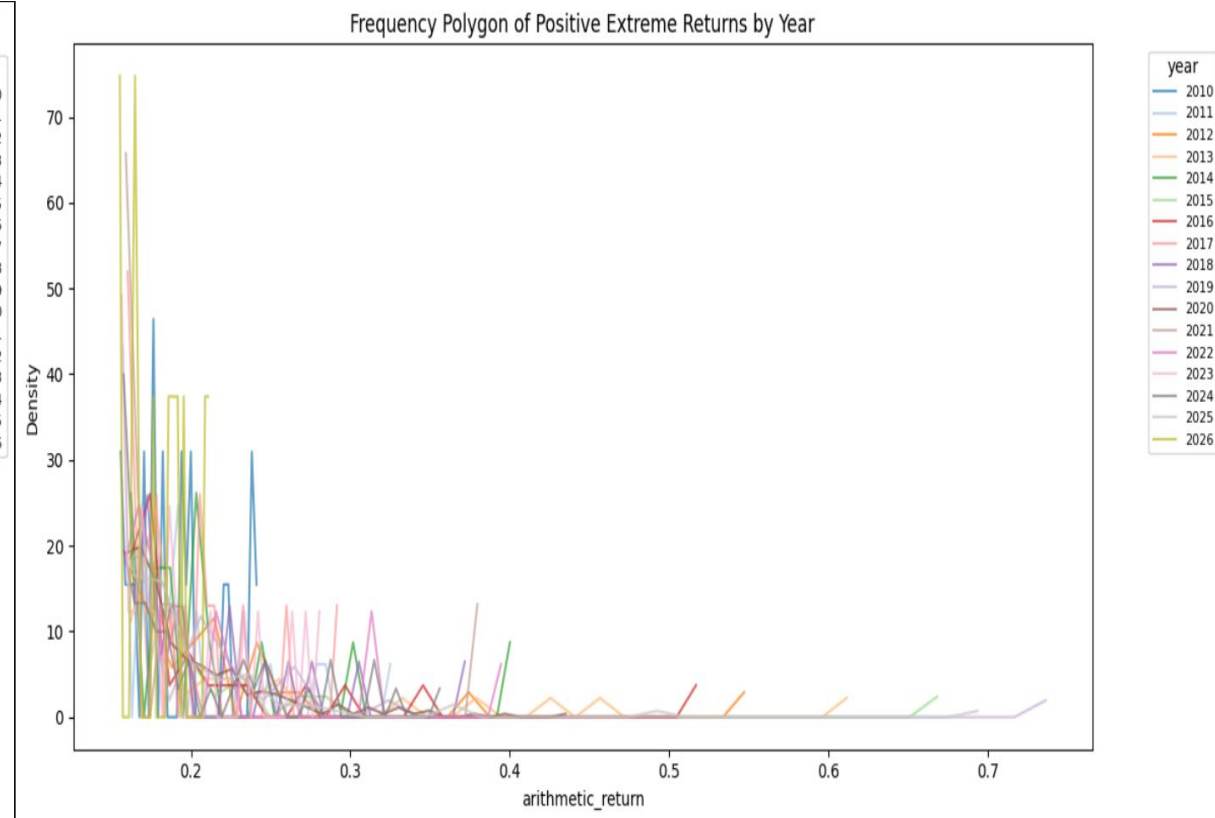
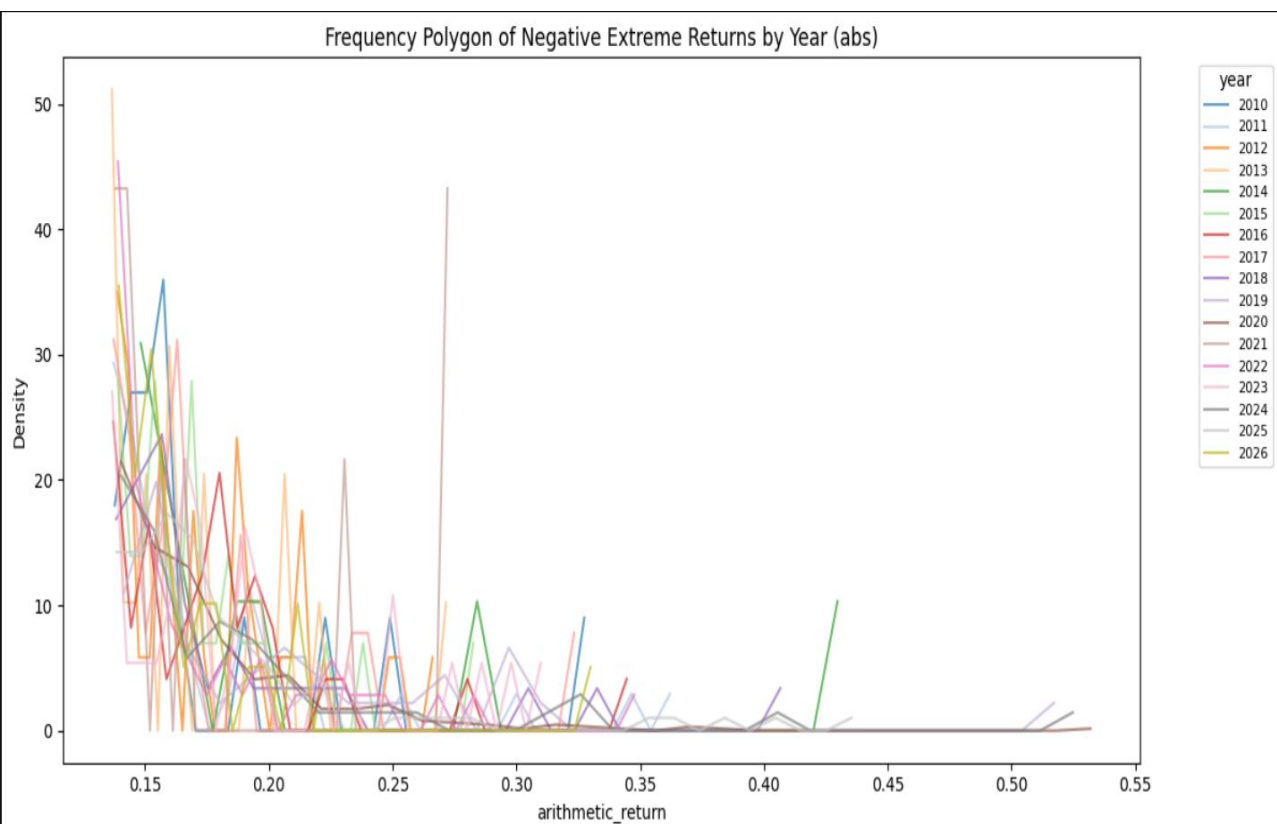


- What years were good / bad for the top 0.1%?

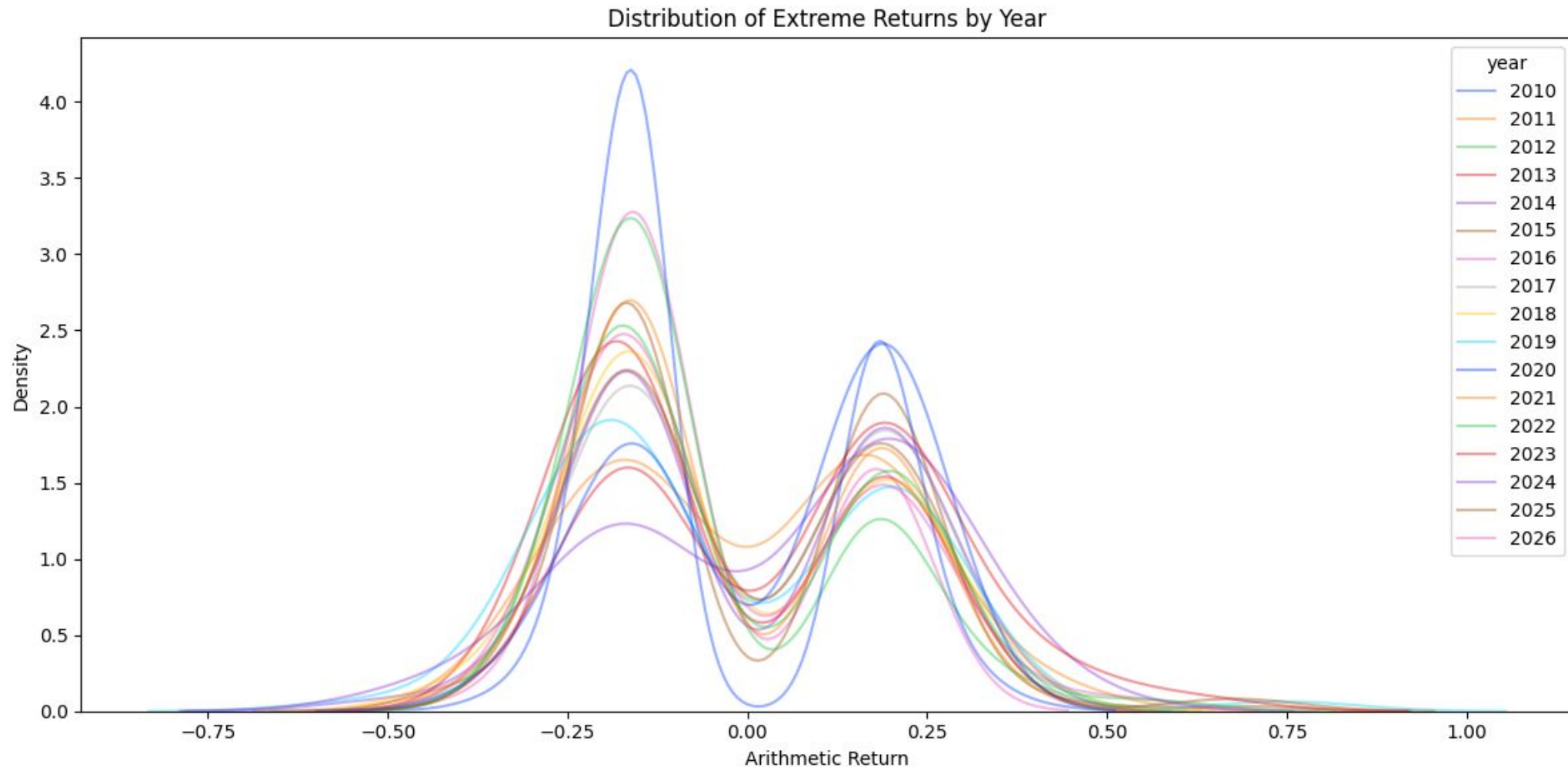
direction	negative_event	positive_event	net	ratio_positive	mean_log_return	median_log_return
year						
2010	17	22	5	0.564103	0.014507	0.146681
2011	44	28	-16	0.388889	-0.050225	-0.149092
2012	39	26	-13	0.400000	-0.038452	-0.153557
2013	21	29	8	0.580000	0.040606	0.153643
2014	10	14	4	0.583333	0.014659	0.151223
2015	29	24	-5	0.452830	-0.018127	-0.147640
2016	34	22	-12	0.392857	-0.044785	-0.153643
2017	20	17	-3	0.459459	-0.021121	-0.144722
2018	32	21	-11	0.396226	-0.048438	-0.155941
2019	35	26	-9	0.426230	-0.049654	-0.150451
2020	470	280	-190	0.373333	-0.053402	-0.153259
2021	10	10	0	0.500000	-0.015867	-0.000785
2022	49	20	-29	0.289855	-0.081468	-0.151663
2023	31	19	-12	0.380000	-0.064066	-0.160101
2024	52	44	-8	0.458333	-0.031347	-0.146023
2025	96	76	-20	0.441860	-0.031597	-0.150361
2026	30	14	-16	0.318182	-0.071460	-0.155533

Most years have more negative events. Most years have a negative mean and median log return.

- What does it look like? (Arithmetic returns to not crush the extremes positives)



- What does it look like? (Arithmetic returns to not crush the extremes positives)



- Do the top 0.1% mean across years are statistically different within each positive extreme and negative extreme? arithmetic_return (left), log_return (right)

Less impact of return definition because we are comparing positive / positive and negative / negative.

Extreme Value Theory and Generalized Pareto Distribution territory.

	tail in arithmetic return	Welch-ANOVA	p_value
0	positive_extremes_across_years	1.461559	0.124434
1	negative_extremes_across_years	0.957364	0.506226

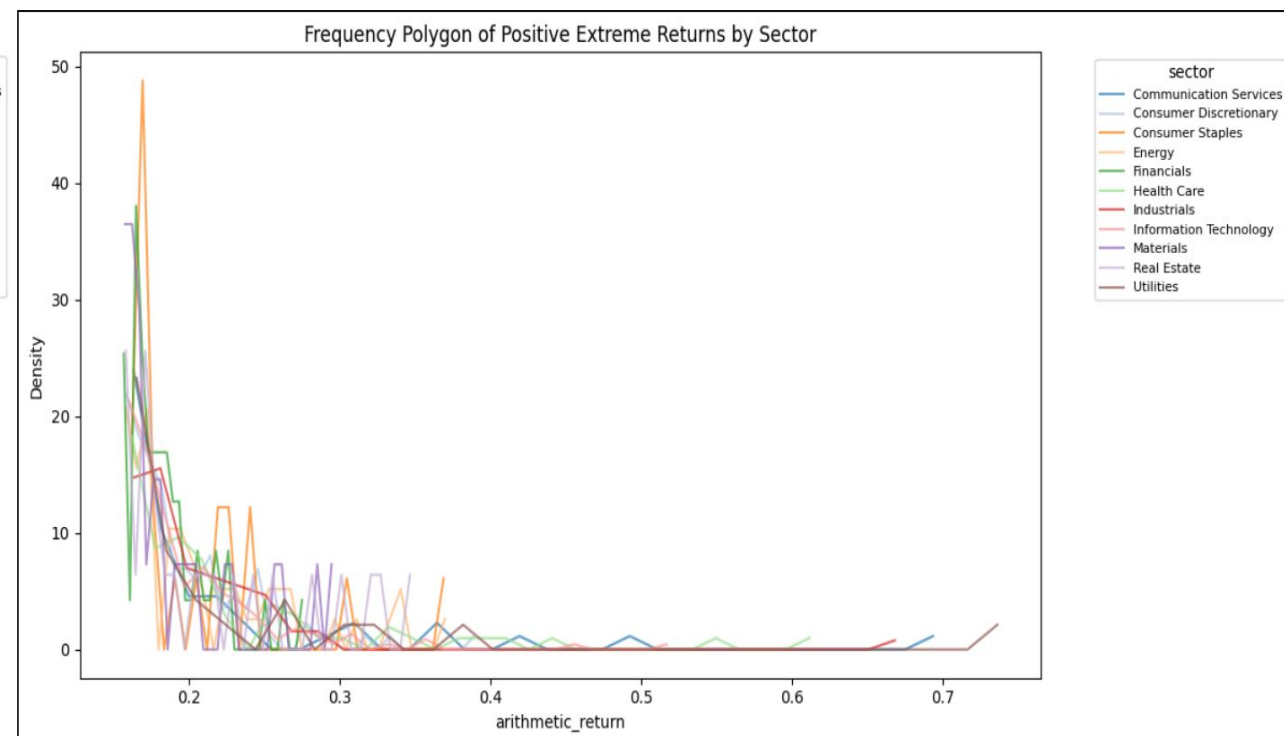
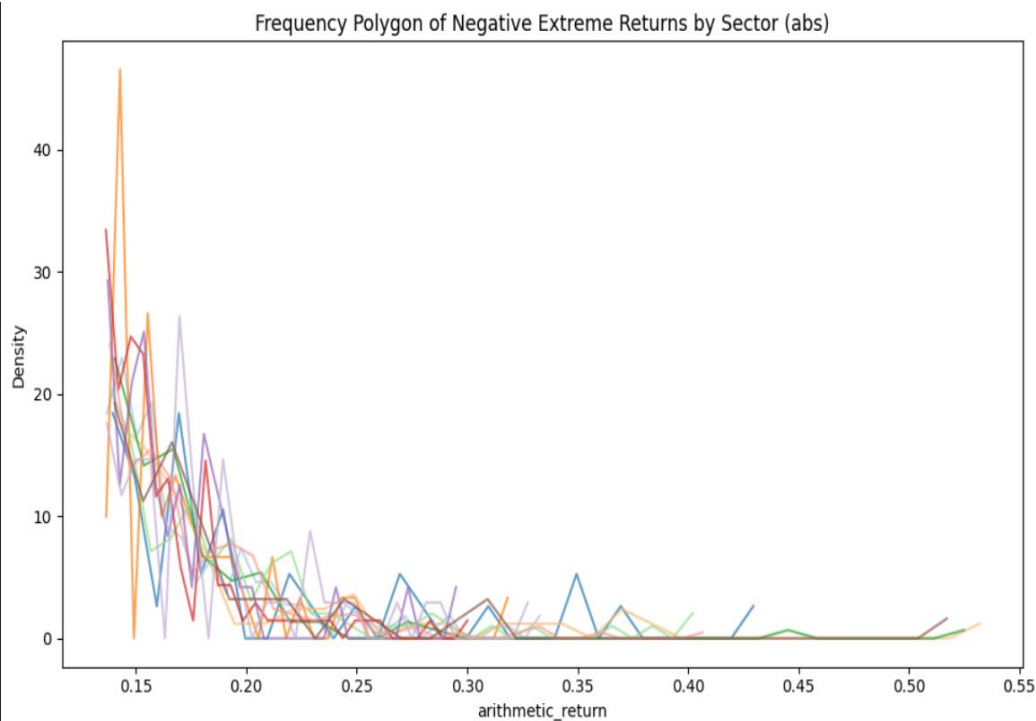
	tail in log return	Welch-ANOVA	p_value
0	positive_extremes_across_years	1.387477	0.158121
1	negative_extremes_across_years	0.924244	0.543047

H0 is not rejected with a $p < 0.05$ significance value. Year over year, the distribution of extreme events does not differ in equality of means.

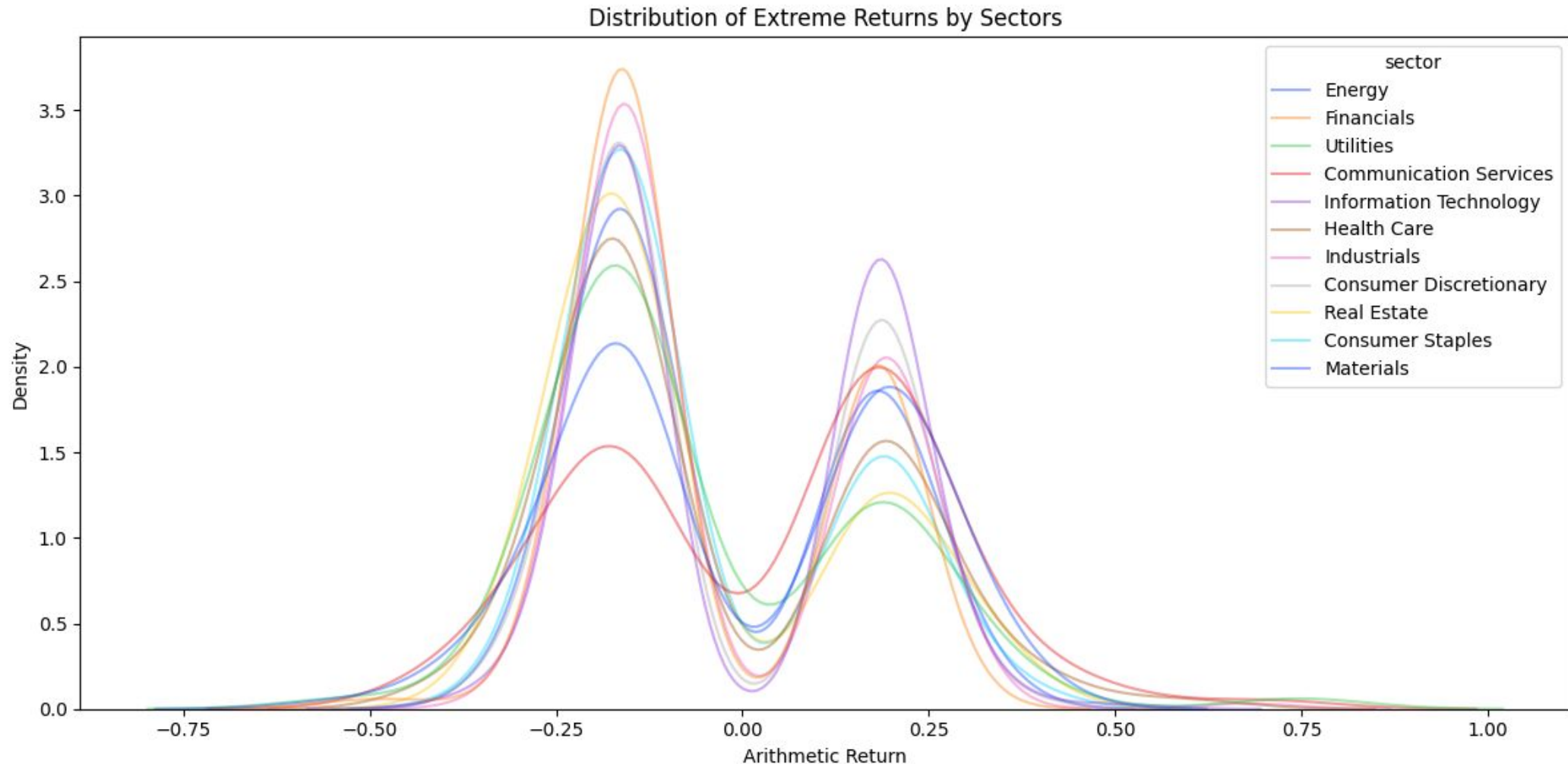
- What sectors performed best in the top 0.1%? Ordered table.

	mean_log_return	median_log_return	positive_ratio
sector			
Communication Services	0.0054	0.1443	0.5581
Information Technology	-0.0251	-0.1477	0.4496
Energy	-0.0333	-0.1482	0.4609
Industrials	-0.0424	-0.1506	0.3776
Materials	-0.0434	-0.1513	0.3973
Consumer Discretionary	-0.0447	-0.1521	0.4037
Health Care	-0.0533	-0.1547	0.3829
Consumer Staples	-0.0664	-0.1562	0.3239
Financials	-0.0678	-0.1535	0.3412
Utilities	-0.0719	-0.1581	0.3333
Real Estate	-0.0782	-0.1656	0.3158

- What does it look like? (Arithmetic returns to not crush the extremes positives)



- What does it look like? (Arithmetic returns to not crush the extremes positives)



- Do the top 0.1% mean across sectors are statistically different within each positive extreme and negative extreme? log_return (right), arithmetic_return (left)

*Less impact of return definition because we are comparing positive / positive and negative / negative.
Extreme Value Theory and Generalized Pareto Distribution territory.*

	tail in arithmetic return	Welch-ANOVA	p_value
0	positive_extremes_across_sectors	2.683449	0.004538
1	negative_extremes_across_sectors	3.445695	0.000281

	tail in log return	Welch-ANOVA	p_value
0	positive_extremes_across_sectors	2.607491	0.005759
1	negative_extremes_across_sectors	3.426325	0.000301

**H0 is rejected with a $p < 0.05$ significance value. Statistical equality of mean is not preserved over sectors.
Most probably because of the Communication Services sector.**

- Do factor models explain extreme returns as well as they explain the full distribution?

$$\text{CAPM : } R_i - R_f = \alpha + \beta(R_m - R_f)$$

R_i — the return of your stock (or portfolio)

R_m — the return of the market (S&P500 index)

R_f — the risk-free rate (DTB3, the 3-month T-bill)

Build a simple OLS for CAPM on AAPL to test.

```

=====
                        OLS Regression Results
=====
Dep. Variable:          stock_excess      R-squared:                0.470
Model:                  OLS              Adj. R-squared:          0.470
Method:                 Least Squares    F-statistic:             3485.
Date:                  Sun, 03 May 2026  Prob (F-statistic):      0.00
Time:                  15:10:02          Log-Likelihood:          11494.
No. Observations:      3936             AIC:                   -2.298e+04
Df Residuals:          3934             BIC:                   -2.297e+04
Df Model:              1
Covariance Type:       nonrobust
=====

```

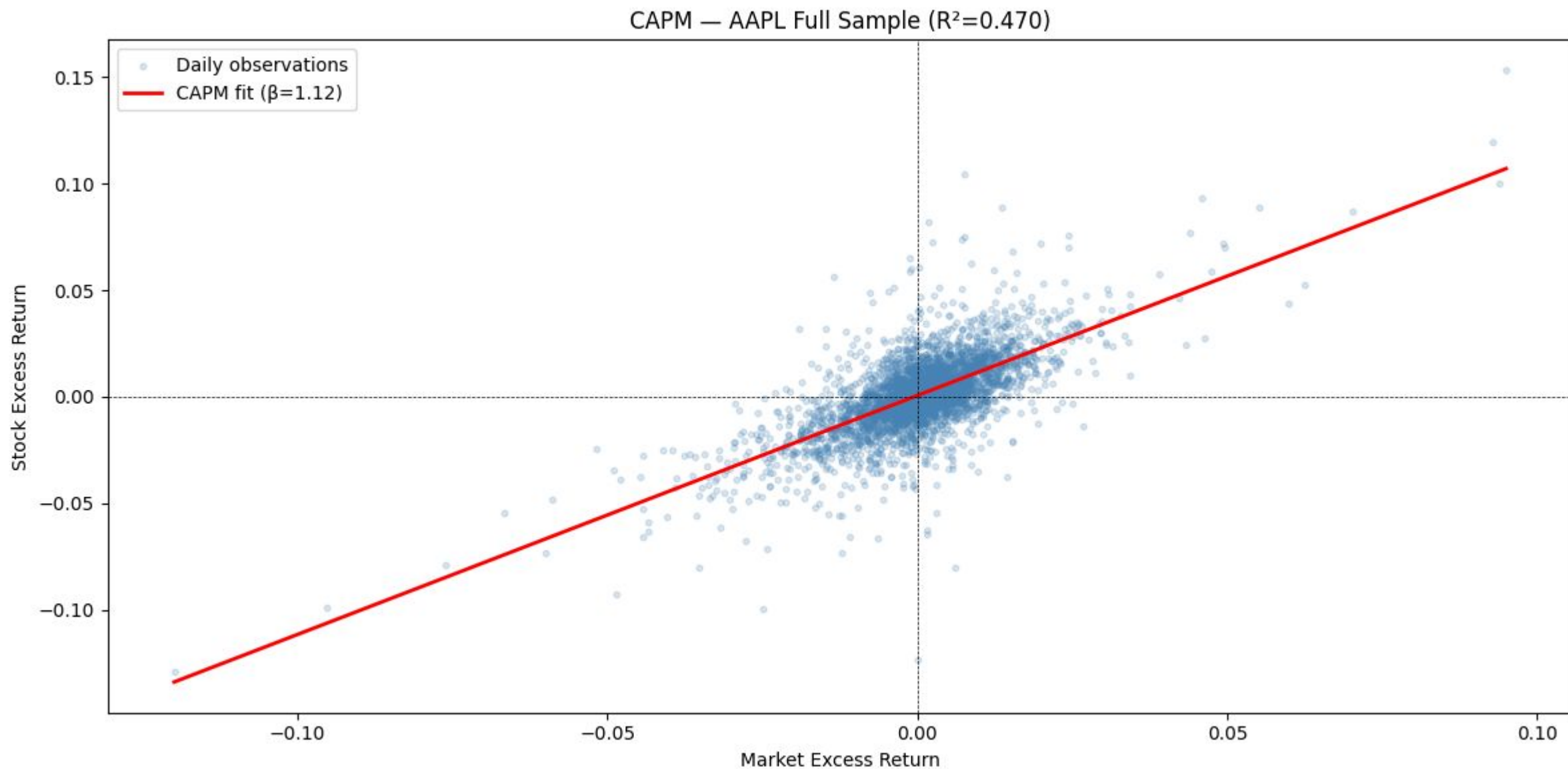
	coef	std err	t	P> t	[0.025	0.975]
const	0.0005	0.000	2.629	0.009	0.000	0.001
market_excess	1.1211	0.019	59.036	0.000	1.084	1.158

```

=====
Omnibus:                551.269      Durbin-Watson:           1.908
Prob(Omnibus):          0.000        Jarque-Bera (JB):        7574.137
Skew:                   0.075        Prob(JB):                0.00
Kurtosis:               9.794        Cond. No.                91.3
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```



Loop on each ticker :

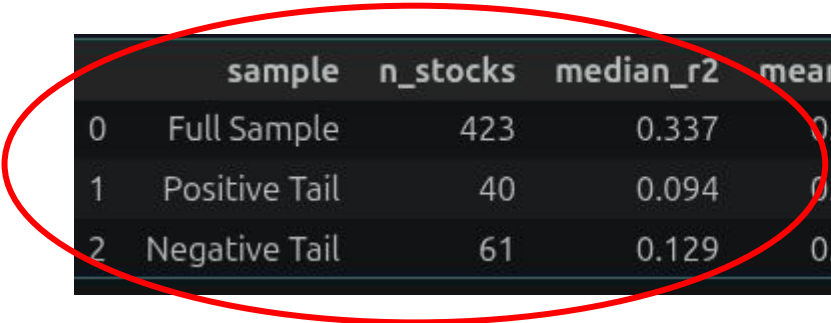
	Ticker	alpha	beta	r2	p_alpha	p_beta	n
0	A	0.000134	1.157895	0.480006	0.523352	0.0	3936
1	AAPL	0.000547	1.121092	0.469760	0.008605	0.0	3936
2	ABT	0.000246	0.738777	0.348994	0.163096	0.0	3927
3	ACGL	0.000352	0.793867	0.334955	0.069585	0.0	3989
4	ACN	0.000133	1.021685	0.515069	0.442290	0.0	3949

Summary :

	metric	mean	median	std	min	max
0	alpha	0.000201	0.000191	0.000237	-0.000457	0.001104
1	beta	0.980084	0.981246	0.283460	0.356161	1.694702
2	r2	0.337979	0.336831	0.118038	0.026121	0.642840

On average, on the S&P500 for the last 15 years, CAPM explains 34% of a stock excess return against market return

Now test the CAPM on the top 0.1%.
Here, separating both tails.



	sample	n_stocks	median_r2	mean_r2	median_beta	median_alpha_ann	pct_sig_alpha	pct_sig_beta
0	Full Sample	423	0.337	0.338	0.981	0.048	9.220	100.000
1	Positive Tail	40	0.094	0.171	-0.065	50.902	100.000	7.500
2	Negative Tail	61	0.129	0.212	0.064	-43.149	95.082	11.475

pct_sig_alpha & pct_sig_beta = percentage of stock where alpha or beta had a significant p-value (>0.05).

CAPM shows very little correlation between stock excess return and market return in the tails compared to the full set !

- Do factor models explain extreme returns as well as they explain the full distribution?

$$FF3 : R_i - R_f = \alpha + \beta(R_m - R_f) + \beta_{SMB} \cdot SMB + \beta_{HML} \cdot HML$$

R_i - the return of your stock (or portfolio)

R_m - the return of the market (S&P500 index)

R_f - the risk-free rate (DTB3, the 3-month T-bill)

SMB - Small minus Big (market cap factors)

HML - High minus Low (book to market ratio)

- FF3 on the whole set of S&P500 from 01.01.2010

```

CAPM median adjusted R² : 0.337
FF3  median adjusted R² : 0.373
Gain                               : 0.036

                                0
sample      FF3 – Full Sample
n_stocks           423
median_r2          0.373295
median_adj_r2      0.372834
mean_r2            0.380349
mean_adj_r2        0.379894
median_beta_mkt     0.9413
median_beta_smb     0.062804
median_beta_hml     0.207891
median_alpha_ann    0.03382
pct_sig_alpha       6.382979
pct_sig_smb         81.08747
pct_sig_hml         90.543735
  
```



- FF3 for the top 0.1%. With adjusted R2.

sample	FF3 – Full Sample	FF3 – Positive Tail	FF3 – Negative Tail
median_r2	0.373295	0.520654	0.581733
median_adj_r2	0.372834	0.072568	0.156952
median_beta_mkt	0.9413	0.216505	0.23334
median_beta_smb	0.062804	-0.311619	-0.006625
median_beta_hml	0.207891	-0.032717	-0.089737
median_alpha_ann	0.03382	49.177992	-42.534714
pct_sig_alpha	6.382979	85.714286	75.409836
pct_sig_smb	81.08747	7.142857	1.639344
pct_sig_hml	90.543735	4.761905	4.918033

Neither CAPM nor FF3 seems good to explain extreme returns. The vast majority of tail behavior remains idiosyncratic and unexplained by systematic factors.

- Do extreme returns have a persistence component to them?
The case for momentum also known as the Carhart 4 factors model.

$$R_i - R_f = \alpha + \beta_m(R_m - R_f) + \beta_{SMB} \cdot SMB + \beta_{HML} \cdot HML + \beta_{UMD} \cdot UMD$$

R_i - the return of your stock (or portfolio)

R_m - the return of the market (S&P500 index)

R_f - the risk-free rate (DTB3, the 3-month T-bill)

SMB - Small minus Big (market cap factors)

HML - High minus Low (book to market ratio)

UMD - Up minus Down (long time winners vs long time losers)

	0	1	2
sample	Carhart — Full Sample	Carhart — Positive Tail	Carhart — Negative Tail
n_stocks	423	42	61
median_r2	0.376237	0.692238	0.854992
median_adj_r2	0.375625	0.137812	0.289059
median_beta_mkt	0.939983	0.522633	0.158503
median_beta_smb	0.053771	-0.085258	0.601842
median_beta_hml	0.196626	0.001649	0.34015
median_beta_umd	-0.032806	-0.292279	1.104322
median_alpha_ann	0.035172	47.957742	-41.121436
pct_sig_alpha	6.146572	50.0	62.295082
pct_sig_umd	59.810875	2.380952	11.47541

Adjusted R2 doubles in the tails !

Rallies go against momentum. Crashes follow it.

The UMD significance tells us that it explains crashes better than rallies.

Does market sentiment explain these extreme returns?
(using a proxy of financial news headlines)

$$R_i - R_f = \alpha + \beta_m(R_m - R_f) + \beta_{SMB} \cdot SMB + \beta_{HML} \cdot HML + \beta_{UMD} \cdot UMD + \beta_{SENT} \cdot SENT$$

R_i - the return of your stock (or portfolio)

R_m - the return of the market (S&P500 index)

R_f - the risk-free rate (DTB3, the 3-month T-bill)

SMB - Small minus Big (market cap factors)

HML - High minus Low (book to market ratio)

UMD - Up minus Down (long time winners vs long time losers)

SENT - minus 1 for low sentiment, plus 1 for high sentiment

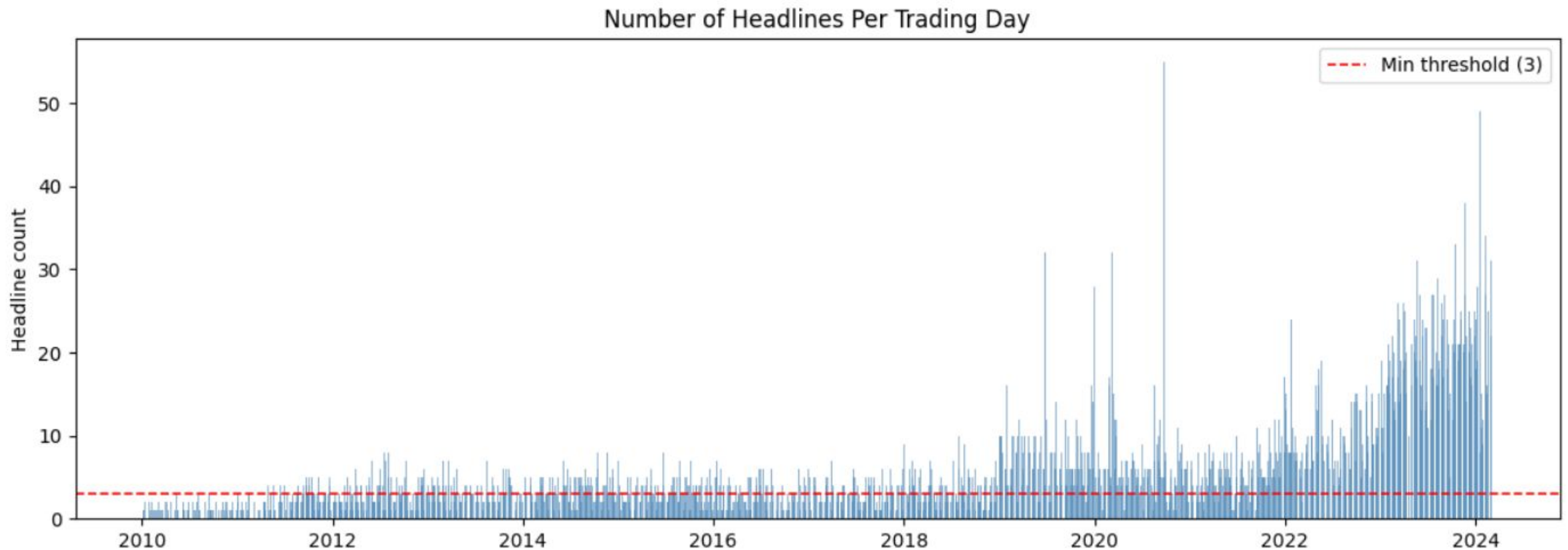
Does market sentiment explain these extreme returns?
(using a proxy of financial news headlines)

Kaggle S&P 500 with Financial News Headlines (2008–2024)

▲ Title	📅 Date	# CP
JPMorgan Predicts 2008 Will Be "Nothing But Net"	2008-01-02	1447.16
Dow Tallies Biggest First-session-of-year Point Drop Ever	2008-01-02	1447.16
2008 predictions for the S&P 500	2008-01-02	1447.16
U.S. Stocks Higher After Economic Data, Monsanto Outlook	2008-01-03	1447.16
U.S. Stocks Climb As Hopes Increase For More Fed Moves	2008-01-07	1416.18

Does market sentiment explain these extreme returns?

But limitations of the dataset... (sparsity and cutoff)



Does market sentiment explain these extreme returns?

FinBERT for NLP analysis :

Hugging Face FinBERT

FinBERT is a pre-trained NLP model to analyze sentiment of financial text. It is built by further training the BERT language model in the finance domain, using a large financial corpus and thereby fine-tuning it for financial sentiment classification. *Financial PhraseBank by Malo et al. (2014)* is used for fine-tuning.

```
Sentiment distribution:  
sentiment_label  
neutral      10816  
negative     4690  
positive     3367
```

Does market sentiment explain these extreme returns?

	0	1	2
sample	Sentiment — Full Sample	Sentiment — Positive Tail	Sentiment — Negative Tail
n_stocks	423	32	47
median_r2	0.413318	0.904962	1.0
median_adj_r2	0.41243	0.259324	0.291162
median_beta_mkt	0.964401	-0.066094	0.087291
median_beta_smb	0.040218	-0.162851	1.164095
median_beta_hml	0.171875	-0.640028	0.197366
median_beta_umd	-0.00604	-0.3372	0.993179
median_beta_sentiment	-0.000361	-0.042487	-0.012072
median_alpha_ann	0.032603	46.67703	-38.152934
pct_sig_alpha	7.565012	31.25	27.659574
pct_sig_sentiment	5.437352	0.0	4.255319

The increase in R-squared with such low sentiment coefficient statistical significance is shaky.
This definition of sentiment does not add any explanatory power.

	Model	Whole Set Median Adj R ²	Positive Tail Median Adj R ²	Negative Tail Median Adj R ²
0	CAPM	0.34	0.09	0.13
1	FF3	0.37	0.07	0.15
2	MOM	0.37	0.14	0.29
3	Sentiment	0.41	0.25	0.29

Overall, the models perform worse in the tails than in the whole set.

- 1) Overall returns on the S&P500 do exhibit fat tail behavior.
- 2) Definition of returns, log vs. arithmetic, start to really matter in the tails.
- 3) Positive tail and negative tail are not symmetrical. Neither in count nor magnitude.
- 4) Extremes distribution seem to be consistent across years.
- 5) Extremes distribution do not seem to be consistent across sectors. (Probably due to the Communication Services outlier. To be tested.)
- 6) Modeling in the tails performs worse than on the full set of returns.
- 7) Factors have very low statistical significance ($p < 0.05$) in the tails overall.
- 8) Crashes, negative tail, seem to be best explained using CAPM and momentum.
- 9) Rallies, positive tail, seem to have very shaky significance with the models used overall.
- 10) I learned about extreme value theory and generalized pareto distributions while making this presentation. So, there is an exploration of these tools to be done next time.



Questions?